

LendSignal User Data Flow

From business onboarding to an ENS-indexed Credit Certificate

The user submits business data, documents and wallet identity. LendSignal processes that information offchain, generates a credit score, updates the CreditCertificateRegistry, and indexes the certificate through ENS records.

1. What The User Fills In

Area	Fields	MVP Handling
Wallet	business wallet, network	Connected wallet becomes credit identity.
Business profile	legal name, country, industry, revenue band, requested loan	Saved in app state/mock backend.
Documents	registration, tax record, statements, invoices, signer auth	Mock upload/checklist; raw files stay offchain.
KYC/KYB	business verified, signer verified	Mock passed/pending/failed status.
ENS	business ENS name	Resolved and later used for certificate discovery.

2. Processing Pipeline

Layer	Input	Output
CRS Adapter	business profile + principal metadata	bureau score, payment risk, fraud risk, CRS report hash
Chainlink AI	documents + profile + prompt	AI score, risk tier, attestation hash, evidence digest
Wallet Analyzer	wallet activity	wallet behavior score and risk flags
Score Combiner	AI 60% + CRS 25% + wallet 15%	combined score and final risk tier

3. Certificate Update

CreditCertificateRegistry receives only public-safe processed information.

Stored Onchain	Never Stored Onchain
business wallet	raw PDFs
combined score and risk tier	full CRS report
certificate status and expiration	bank statements or tax records
attestation hash, CRS hash, evidence digest	KYC/KYB documents or personal identity files

4. ENS Indexing

ENS is the discovery layer. CreditCertificateRegistry remains the source of truth.

ENS Text Record	Value
lendsignal.registry	<CreditCertificateRegistry address>
lendsignal.certificate	<certificate id>
lendsignal.attestation	<Chainlink attestation hash>
lendsignal.crs-report	<CRS report hash>
lendsignal.risk-tier	<risk tier>
lendsignal.agent	<LendSignal agent ENS name>

5. Lending Decision

Check	Pass Condition
ENS resolution + records	Name resolves to borrower wallet and points to registry/certificate.
Certificate	Active and not expired.
Score	Combined score above lending threshold.
Risk tier	Low or medium default risk.
Liquidity	Vault has enough available funds.

6. Update Triggers

New documents, refreshed CRS report, re-run Chainlink AI review, wallet behavior changes, repayment/default events, or KYC/KYB status changes.

Hackathon Version

Use mocked documents, mocked KYC/KYB, CRS_USE MOCK=true, and real or mocked Chainlink AI depending on API key availability. The demo should still update CreditCertificateRegistry and show ENS records pointing to the active certificate.